TO: POUGHKEEPSIE

FROM: POUGHKEEPSIE
Dept. 539, Lab. Annex

MEMO TO: Dr. W. Buchholz

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SUBJECT: Manual Describing the 3-1 Computer System

In the preparation of the 3-1 computer system manual, the several items herein listed should be considered for inclusion in sufficient detail so that design groups can study methods of realization.

1. Full Indicator List
   a. including description of individual indicator behavior (i. e., cumulative, independent),
   b. alternative interpretations of indicators if any exist.
2. Index adder operation.
   a. What is an illegal address? What is done if an illegal address is identified?
   b. Is memory address wrap-around an acceptable mode of operation?
   c. Is recomplementation always specified?
   d. Is a resultant negative effective address acceptable?
3. Internal Clock.
   a. What time measurements are to be provided?
   b. What output does the clock provide, where is it accessable?
   c. How is the clock initiated or set?
   d. Can the clock interrupt the computer of its own volition?
   e. Are special "clock" instructions required?
4. Inter-computer Communication.
   a. How is inter-computer communication realized?
   b. What devices, units, etc., are required, permitted or affected?
   c. Are any special instructions required?
5. Actual indexing abilities.
   a. How many indexing techniques are to be provided?
   b. What limitations, if any, are imposed on the indirect address chaining index? (i. e., an indirect address leads only to a direct address, or an indirect address can lead indefinitely to other indexable indirect addresses.)

6. Presently unlisted instructions.
   a. Replace (4 var.)
   b. Execute
   c. Swap (re-definition required)
   d. Special Editing instructions

7. Card image conversion.
   a. Is card image conversion to be accomplished in the computer unit or in the I/O unit?
   b. Are any special machine capabilities to be defined for this purpose?

8. Table Look-up.
   a. Are any special 'B' machine (exclude Harvest difinition from this section) capabilities to be defined for this purpose?
   b. Any special instructions required?

9. Operators Console (also Maintenance Console).
   a. Is an operators console to be provided?
   b. How will the console connect to the computer? (i. e., physical entity, low speed channel, other).
   c. What manipulative abilities will a console provide an operator?
   d. What information will the console present to the operator? (i. e., physically via lights, indirectly via typewriter.)

10. Editing.
   a. Are any special instructions or machine features to be defined to facilitate editing?

11. Harvest (also Sigma where applicable).
   a. What is the latest up-dated list of special Harvest instructions?
   b. Word and control formats?
   c. Special interlocking and/or indicator criteria?
   d. Is 3-1 compatibility maintained?

12. Multiprogramming.
   a. Which technique or techniques of multiprogramming are being advocated, and do these require any machine features or capabilities not already defined?

13. High-Speed Exchange (also Low-Speed Exchange).

a. Even though these respective units remain mostly undefined, what are the implications of operation pertinent for inclusion in the overall philosophy of the computers proper?

W. Wolensky.

WW/pkb

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